

Football Research Series — Final Summary

Europe's top five leagues · Studies 1–5 · status as of 6 October 2026

1. The short version

Five studies on one dataset — every player-match in the Premier League, La Liga, Serie A, Bundesliga and Ligue 1 from 2015/16 to 2024/25 (525,328 player-matches), extended weekly with 2025/26 and 2026/27. Each study asks how the **context** a player plays in changes his numbers, and each builds on the one before.

Study	Question	Answer
1 · Weather & crowds	Do weather and crowds change player performance?	Weather barely matters (rain, wind, humidity ruled out beyond $\pm 2\text{--}3\%$; temperature is the one open lead). Empty stadiums roughly halved home advantage , and it came back with the fans.
2 · Home advantage	Which players benefit most from playing at home?	Everyone, by about the same amount: +27% xG per minute at home for the same player. No player or club has a reliable personal home edge (0 of 4,665 players).
3 · Opposition	How much does opponent quality distort player stats?	A lot, and equally for everyone: –14% xG per +1 SD of opponent strength , about a third less against the strongest fifth than the weakest. No flat-track bullies, no big-game players.
4 · Similarity	Can we find functionally similar players across leagues?	Yes, within limits: context-adjusted profiles find sensible matches (Bruno → Serie A: Samardžić, Dybala, Chukwueze; nobody matches his chance creation). But similarity is not a forecast .
5 · Transferability	How much performance	Most of it on average (97% vs

Study	Question	Answer
	survives a league move, and can we predict it?	comparable stayers), but one league-difficulty ladder decides the direction: into the Premier League keeps 82%, out of it 116% . A model of the player's own history, age and clubs beats "he'll do what he did" on unseen seasons; similarity adds nothing to it.

The five lessons that run through everything

1. **Context moves the numbers, and it moves everyone the same.** Home, crowd and opposition effects are large, but they belong to the match and the league, not to individual players or clubs. Every "special" player we looked for — home specialists, big-game players, crowd-sensitive players — disappeared once noise was separated from signal.
2. **Most apparent individual differences are small samples.** A player's home/away or strong/weak-opponent split is 5–8% signal; raw rankings of these traits are regression to the mean in action (Study 2: season-to-season correlation 0.018).
3. **A player's overall level is the best predictor of almost everything.** His overall opponent-adjusted output predicts his output against elite teams better than his record against elite teams (Study 3); his own pre-move history predicts his post-move output better than similar players do (Studies 4 and 5).
4. **Similarity is a discovery tool, not a forecasting tool.** Comparable players help find candidates; they add nothing to a forecast once the player's own history is known (Study 5, critical experiment).
5. **Leakage discipline matters.** Understat's match "forecast" turned out to be post-match; it was never used. Opponent strength is built only from pre-match odds; every Study 5 prediction uses only information available at the decision date, and the final model was scored once on locked seasons.

2. The foundation: data and methods

Data

- **Coverage:** 18,011 matches (18,008 with usable player data), 160 teams, 8,348 players, 525,328 player-matches, ~452,000 shots; five leagues, 2015/16–2024/25 (the frozen study data). Live seasons 2025/26 (complete: 1,752 matches) and 2026/27 (in progress) are added weekly and never alter frozen results.
- **Per player-match:** minutes, goals, assists, shots, xG, xA, key passes, xGChain, xGBuildup, cards, granular position, home/away, opponent, kickoff. **Per shot:** location, xG, situation, body part, last action, assister.
- **Sources:** Understat (player and shot data), football-data.co.uk (pre-match Pinnacle odds; market-average closing odds for live seasons), fixturedownload.com and openfootball (kickoffs), Wikipedia/Wikidata and OpenStreetMap (stadiums), Open-Meteo ERA5

(weather), dated crowd-restriction calendars per league, and — for Study 5 — the CC0 Transfermarkt-derived dataset (dcaribou/transfermarkt-datasets) for dates of birth.

- **Not in the data:** defensive actions (tackles, pressures), passes completed, carries, possession, attendance figures, injuries, fees, contracts. This is why the studies focus on attacking output and exclude goalkeepers.

Key methods

- **Within-player comparisons:** Poisson pseudo-likelihood (PPML) on per-minute output with player × team × season fixed effects, so each player is compared with himself; errors clustered by match.
- **Signal vs noise:** individual effects get calibrated standard errors (pseudo-splits or shuffles within each player's own matches), DerSimonian–Laird between-player spread and empirical-Bayes shrinkage; persistence measured against shuffled data.
- **Opponent strength:** a sequential market rating from pre-match odds (each match uses only earlier matches), checked against an xG rating, Elo, opening odds and points per game; 0 changed ratings in 155,282 leakage checks.
- **Multiple testing:** Benjamini–Hochberg corrections and permutation tests; placebo tests (e.g. closed-doors dates shifted 2–4 years) for every headline effect.

3. Study 1 — Weather, crowds and player performance

Status: crowd results final; weather results final for four leagues. The five-league weather confirmation (with Ligue 1) and the final hand-written weather conclusion and freeze are **still pending** — the planned overnight run did not produce its outputs.

Weather

- **Weather barely changes how much players produce.** EPL over ten seasons: 0 of 20 weather × metric effects survive correction. La Liga + Serie A + Bundesliga: rain, wind and humidity effects within ±2.5%.
- **Temperature is the one open lead:** per +10 °C, key passes +3.3% (three-league pool), concentrated in Serie A (xA +10.7%, key passes +5.8%). Part of it is "warm stadiums" rather than warm days (shrinks with a stadium control). Not yet a finding.
- **No effect on specific actions:** six pre-registered hypotheses (rain and long shots, wind and headers/crosses, heat and late-game output) — 0 of 6 survive.
- A promising EPL 2024/25 lead (forwards' xA +29% in rain) failed to replicate on three other leagues (−1.4%).

Crowds

Home ÷ away (team level)	With fans	Behind closed doors
Goals	1.27	1.11
xG	1.29	1.13

Home ÷ away (team level)	With fans	Behind closed doors
Shots	1.23	1.13
Yellow cards	0.89 (home side booked less)	1.02 (advantage gone)

Every league lost xG home advantage without fans (Ligue 1 lost its whole goals edge) and it recovered when crowds returned (pooled xG edge +30% → +13% → +26%).

4. Study 2 — Individual home advantage

Status: complete.

- **Everyone does better at home by about the same amount.** Same player, same team and season: xG +27%, xA +25%, shots +22%, key passes +21%, goals +24% per minute.
- **The crowd carries most of it.** Behind closed doors the edge fell to about +8–14% for every metric and came back with fans (xG +28% → +12% → +25%). The betting market priced the same loss (home/away win-probability ratio 1.58 → 1.31 for evenly matched teams).
- **Leagues differ:** La Liga largest (xG +35%), Serie A smallest (+23%).
- **No individual home specialists.** Each player's estimate is only 5–8% signal; **0 of 4,665 players** have a reliable personal home edge; within a team-season, individual differences vanish for xG, shots, key passes and goals. The 2015–19 "top 10 home players" were exactly average in 2022–25 (correlation between periods 0.018).
- **No club home edge either:** against their own league's average, clubs differ by about ±1% (0 of 121 clubs reliably different); a club's past home edge does not predict its future one.

5. Study 3 — The opposition effect

Status: complete (six milestones; every robustness test holds).

- **Opposition quality is one of the biggest contexts in the numbers.** Per +1 SD of opponent strength (same player, team and season): xG −13.8%, xA −15.1%, shots −10.5%, key passes −11.4%, goals −15.8%. Strongest vs weakest fifth of opponents: about a third less (xG −33%, xA −35%).
- **Same for everyone:** the same across positions and home/away; unchanged behind closed doors (unlike home advantage). **0 of 23,022** player × metric estimates show a reliable individual "resistance"; season-to-season correlation 0.012. The raw data's ~29% "flat-track bullies" and ~26% "big-game players" are exactly what noise produces.
- **Leagues differ:** steepest in Serie A (xG −16%, xA −18% per SD); shot and key-pass volume barely drops against strong sides in La Liga and Ligue 1 (about −7%).
- **Practical rule:** to predict a player's output against elite opponents, use his overall opponent-adjusted level, not his record against elite teams (xG r 0.84 vs 0.75).
- **Season adjustments are small** (typically ±2%; up to ±10% for players used unevenly, e.g. Callum Wilson 2023/24 −9.9%) because everyone plays everyone twice. Adjustment matters for partial seasons, single matches and cross-league comparisons. Bruno Fernandes's elite chance creation survives adjustment (0.467 → 0.460 xA per 90 in 2022/23, 99.6th percentile).

6. Study 4 — Cross-league player similarity ("Finding the next Bruno")

Status: complete (eight milestones; interactive search on the website).

- **Roles from the data:** clustering Understat's granular positions on per-90 output gives seven roles; a central attacking midfielder produces like a wide forward, not like other central midfielders.
- **Profile:** nine metrics — xA, key passes, xGChain, xGBuildup, non-penalty xG, shots per 90, plus headed share, in-box share and xG per shot — adjusted for opponent, venue and league environment, z-scored within role; cosine similarity.
- **Validation without opinions (self-retrieval):** three-season profiles find the same player in the top 9% of a destination league's role pool after a league move; adjusted inputs beat raw ones for players who changed league; single seasons are noisy profiles.
- **Bruno → Serie A:** robust matches Samardžić, Dybala and Chukwueze — but every one creates less: Bruno's chance creation is 3.6 SD above his role average; Serie A's best is 1.9. **Statistically there is no like-for-like Bruno in Serie A.**
- **The limit that set up Study 5:** after a real move, the player's own pre-move profile forecasts his new-league output as well as his comparables do; about half of a player's distinctiveness survives a move.

7. Study 5 — Transferability ("Will his game travel?")

Status: complete (nine milestones; website section with a transfer calculator).

Population

Understat has no transfer records, so moves were inferred from club changes (including deadline-day moves after early matches, e.g. Ronaldo 2021). Of 2,917 cross-league moves, the primary population is **428 summer moves** (2016/17–2025/26) by forwards and midfielders with ≥ 900 minutes before and after; every exclusion is listed with its reason. Dates of birth were linked from the Transfermarkt-derived dataset by matching the same club, date, minutes and goals (100% of the 428 moves).

How much survives

Retention = output after $\div$ output before (xG + xA per 90). Because output drops after any strong season (regression to the mean), every group is also compared with **players who stayed at their club** in the same role with the same output and age.

xG + xA kept vs comparable stayers	Moving into	Moving out of
Premier League	82% (76–87%)	116% (106–126%)
La Liga	100%	100%

xG + xA kept vs comparable stayers	Moving into	Moving out of
Serie A	96%	99%
Bundesliga	119% (107–130%)	90%
Ligue 1	118% (108–129%)	90%

- **On average a move costs ~3%** (97%, CI 93–101%), but **one ladder of league difficulty** explains all 20 directions (pair-specific effects add nothing, $p = 0.91$): Premier League hardest, Serie A and La Liga in the middle, Bundesliga and Ligue 1 easiest. Serie A → Premier League keeps ~80%, Premier League → Serie A ~114%.
- **Younger players keep more**, beyond normal ageing: about –2% per year of age (U21 109%, 30+ 93%).
- **The destination club matters as much as the league**: weakest third of destination clubs 79%, strongest third 106%; a step up in club strength keeps 114%, a step down 81%.
- **Players coming off a jump in output drop more** (–9% per SD of last-season change).
- **No effect**: role (once regression to the mean is removed), consistency over three seasons, profile distinctiveness, penalty duties, loans.

Can it be predicted?

Model (first season after the move)	Error (MAE), locked test 2023/24–2025/26	R ²
"He'll do what he did before" (post = pre)	0.129	0.58
Pull toward the role average	0.117	0.65
Context (age, minutes, both clubs' strength, leagues)	0.102	0.74
Final model: context + 3-season history	0.096	0.76

- The final model was fixed before the test seasons were opened and scored once on 142 unseen moves; it beat the naive baseline by 0.033 xG + xA per 90 (CI 0.018–0.049) and did slightly better on the unseen seasons than on earlier ones (no overfitting). Robust to minutes thresholds, loans, refitting each season and an alternative target (xGChain).
- **The critical experiment**: adding Study 4's similarity — what the most similar destination-league players produce, match quality, distinctiveness, how similar movers fared — changed the error by 0.000. Comparables alone were worse than the player's own history. **Similarity finds candidates; it does not forecast them.**
- **Chance of keeping ≥ 75%**: ranks players well (AUC 0.80) and beats a same-for-everyone base rate (Brier 0.179 vs 0.224); calibrated at the top. Lower thresholds (50%, 60%) were not well calibrated and are not used. Individual forecasts are uncertain: a typical 80% range is ± 0.2 xG + xA per 90.

- **Known weakness:** the league effect is a fixed amount, not a percentage, so low-output central midfielders moving into the Premier League are under-predicted by about 0.06 (Ugarte). A percentage-based league effect is the next version.

Case study — Manchester United, summer 2024 (leakage-free)

Standing at 1 June 2024 with only what was known then, every striker and central midfielder in the other four leagues was ranked. **Zirkzee:** predicted 0.46 xG + xA per 90 at United, actual 0.47 — right number, but 48th of 106 strikers by expected output (the board favoured Boniface, Guirassy, Dovbyk, Openda, Osimhen). **Ugarte:** predicted 0.02, actual 0.14 — a miss on the model's known weak spot, for a player bought for defensive work this data cannot see. Candidates who did join Premier League clubs: strikers 4 of 4 inside their 80% range, midfielders 5 of 6.

8. Live tracker — do the findings hold on new seasons?

Every Monday the pipeline adds the weekend's matches and re-fits Studies 2 and 3 on seasons the studies never saw.

- **Home edge, 2025/26:** xG +27.1% (frozen estimate +26.6%) — holds. 2026/27 so far: +21.0% (wide interval, early season).
- **Opposition effect, 2025/26:** xG -11.4% per SD (frozen -13.8%) — same direction and size. 2026/27 so far: -12.9%.
- **Predictions confirmed on 2025/26:** a player's past raw home edge says nothing about this season ($r = 0.011$; the league average is the better forecast); a player's overall adjusted level beats his past record against elite teams for every metric (xG 0.76 vs 0.68).

9. How the studies fit together

Study 1 asked whether the environment (weather, crowds) moves performance — weather barely, crowds clearly. Study 2 took the crowd effect to the individual: home advantage is real, large and shared by everyone. Study 3 did the same for opposition: large, shared, with no individual "big-game" trait. Together they gave clean, context-adjusted player numbers. Study 4 used those numbers to find similar players across leagues, and found the limit: similarity describes, it does not forecast. Study 5 tested forecasting directly on real transfers and built a validated, leakage-free model of how much output survives a move — in which similarity had to earn its place and did not.

10. What we have built

- **Website** ("Home Turf & Hard Opponents"): <https://claude.ai/artifact/VKIERFFGQaBpxDJoopCikR> — sections for the live tracker and Studies 2–5, including the Study 4 similarity search (any player → any league) and the Study 5 explorer, league matrix, Manchester United case study and **transfer calculator** (any forward or midfielder → league → club → expected xG + xA per 90, 80% range, chance of keeping $\geq 75\%$). Rebuilt weekly; republished with /publish-site.

- **Weekly pipeline** (Monday 07:00, Mac launchd): Understat and odds fetch → kickoff reconciliation → live tables and ratings → tracker → Study 4 profile refresh → Google Sheets sync → site rebuild.
- **Google Sheets** in this account (folder "Football research"): one spreadsheet per league (players and shots per season), shared tables, study results and findings.
- **Key data tables**: player-match and shot tables (frozen + live); team strength ratings; player home advantage; opponent-adjusted player performance; opposition resistance; Study 4 similarity profiles; Study 5 transfer list, features, predictions and the 2024 recruitment board.
- **Code and documents**: project at ~/dev/weather_football — one PRD per study (docs/), code per study (src/study2 ... src/study5), milestone reports (outputs/), findings files (findings/).

11. Limits and open items

- Observational data: effects are associations within carefully matched comparisons.
- Output only — no defending, pace, possession, injuries, fees or contracts; goalkeepers and (in Study 5) defenders excluded.
- The closed-doors period is mostly one season, so per-player crowd estimates are noisy; no attendance figures.
- Cup matches and fixture congestion are not in the data.
- **Open**: Study 1's five-league weather confirmation, final weather conclusion and freeze; Study 4's user-curated "known similar pairs" check; Study 5's percentage-based league effect, to be validated on moves after 2025/26.

12. Corrections made along the way

- Understat's win/draw/loss "forecast" was at first described as pre-match; it is post-match ($r = 0.96$ with the match's xG difference). It was never used in any model.
- Study 2's "stable part" magnitudes are upper bounds: the cross-season covariance estimator is biased upward even with no true persistence (found in Study 3). Conclusions unchanged.
- Study 5: a promoted-club term was first read as an imputation artefact; it is a real effect (players joining promoted clubs produce more than the club's rating predicts). The early "centre-forwards drop most" reading was mostly regression to the mean (96% vs stayers, no clear role effect).